

TOKKO, Russian Federation

WMO: 311370

Lat: 56.283N

Lon: 131.133E

Elev: 850

StdP: 91.53

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.8	-48.9	-52.1	0.0	-50.5	-50.2	0.0	-48.8	5.1	-19.0	3.4	-27.8	0.0	50	0.758

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	27.2	17.2	25.1	16.5	23.0	15.4	18.5	24.7	17.3	23.1	16.3	21.6	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	12.9	20.7	15.2	11.9	19.8	14.0	11.1	18.8	56.1	24.8	52.3	23.1	49.0	21.8	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.8	5.0	3.5	-53.1	29.7	2.8	2.1	-55.2	31.1	-56.8	32.3	-58.4	33.5	-60.5	35.0		
(5)			-52.9	20.8	2.9	1.9	-55.0	22.1	-56.7	23.3	-58.3	24.3	-60.4	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.9	-36.1	-31.3	-21.2	-7.5	4.2	11.7	14.5	11.5	3.9	-8.5	-26.0	-35.8		
	DBStd	19.34	6.25	5.78	6.56	6.17	4.15	3.38	3.07	3.39	4.41	6.34	8.24	6.13		
	HDD10.0	7572	1429	1156	969	524	183	21	3	24	187	575	1081	1420		
	HDD18.3	10327	1688	1389	1227	774	437	201	122	214	433	833	1331	1678		
	CDD10.0	291	0	0	0	0	4	70	143	69	4	0	0	0		
	CDD18.3	5	0	0	0	0	0	1	4	1	0	0	0	0		
	CDH23.3	352	0	0	0	0	3	72	218	58	1	0	0	0		
	CDH26.7	61	0	0	0	0	0	9	46	6	0	0	0	0		
	WSAvg	0.8	0.3	0.5	0.9	1.2	1.5	1.2	1.0	1.0	1.1	0.8	0.4	0.2		
(15) Precipitation	PrecAvg	473	10	8	9	27	48	68	86	89	59	36	17	12		
	PrecMax	658	23	17	28	66	115	114	214	181	127	86	33	28		
	PrecMin	310	2	2	1	2	7	18	27	23	10	9	3	4		
	PrecStd	77	5	4	6	16	24	24	40	41	28	18	8	6		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-13.7	-10.0	1.5	12.1	22.6	28.1	31.0	27.7	21.0	9.2	-1.8	-14.5		
		MCWB	-14.5	-11.2	-1.4	6.1	12.1	16.3	19.0	17.7	12.9	4.8	-3.7	-15.1		
	2%	DB	-19.2	-13.9	-2.6	8.1	19.2	25.8	28.0	25.2	17.6	5.5	-5.9	-19.2		
		MCWB	-19.5	-14.8	-4.8	3.4	10.5	15.9	17.9	17.0	11.3	2.4	-6.8	-19.5		
	5%	DB	-22.3	-17.6	-5.3	5.3	16.3	23.6	26.1	23.0	15.1	2.7	-9.3	-22.8		
		MCWB	-22.6	-18.2	-7.1	1.2	9.0	14.6	17.1	16.0	10.1	0.5	-9.9	-23.0		
	10%	DB	-25.5	-20.7	-8.2	3.1	13.7	21.5	23.8	20.6	12.7	0.7	-12.3	-25.5		
		MCWB	-25.7	-21.1	-9.7	-0.3	7.6	13.4	16.4	14.9	8.7	-0.8	-12.9	-25.7		
	0.4%	WB	-14.4	-11.2	-1.2	6.5	12.8	18.3	20.6	19.4	14.5	5.2	-3.3	-15.1		
		MCDB	-13.5	-10.1	1.1	11.4	20.6	24.7	27.3	25.5	18.9	8.7	-1.9	-14.7		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-19.4	-14.8	-4.6	3.5	11.0	16.7	19.0	17.9	12.3	2.7	-6.8	-19.5		
		MCDB	-19.1	-13.9	-2.6	7.7	18.1	23.8	25.8	23.5	16.2	5.2	-5.8	-19.2		
	5%	WB	-22.6	-18.2	-7.0	1.5	9.5	15.5	17.9	16.6	10.4	0.8	-9.9	-23.0		
		MCDB	-22.3	-17.7	-5.2	4.8	15.3	21.7	23.9	21.5	14.2	2.4	-9.3	-22.8		
	10%	WB	-25.7	-21.1	-9.6	-0.1	8.0	14.3	16.9	15.4	9.0	-0.8	-12.8	-25.7		
		MCDB	-25.5	-20.7	-8.2	2.8	13.2	20.1	22.4	19.9	12.2	0.6	-12.3	-25.5		
	5% DB	MDBR	14.0	18.5	20.3	16.2	13.1	15.4	15.1	14.9	12.8	12.6	14.9	12.2		
		MCDBR	15.0	19.7	21.0	17.3	19.8	21.3	21.2	20.6	18.2	13.7	14.7	12.6		
		MCWBR	14.5	18.8	18.6	12.7	12.0	12.3	12.0	12.7	12.1	10.6	13.6	12.2		
		MCWBR	14.9	19.6	20.2	15.3	17.9	18.0	17.2	17.6	16.3	12.8	14.5	12.5		
(40) Clear-Sky Solar Irradiance	taub		0.227	0.257	0.300	0.386	0.430	0.420	0.436	0.376	0.325	0.293	0.239	0.216		
		taud	2.101	2.110	2.034	1.869	2.022	2.213	2.194	2.368	2.430	2.254	2.088	2.035		
	Ebn at Noon		685	807	850	814	805	823	798	832	834	753	646	583		
		Edh at Noon	57	87	123	171	157	132	132	104	84	75	57	46		
(44) All-Sky Solar Radiation	RadAvg		0.92	2.03	3.71	5.17	5.07	5.45	5.00	4.00	2.73	1.75	1.10	0.63		
	RadStd		0.05	0.10	0.17	0.31	0.46	0.40	0.44	0.38	0.28	0.19	0.06	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (2 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page